AV WATCH

Mobile App Product Requirements Document

A Transparency Platform for Autonomous Vehicle Accountability

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Team	Noah Baier, Evan Haas, Joshua Mussman, Monica Paz Parra
Advisor	Morgan Ames
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1. Executive Summary

AV Watch is a mobile-first transparency platform that empowers citizens, researchers, and policymakers to report, track, and analyze autonomous vehicle (AV) incidents on public roads. As AV deployments expand rapidly across U.S. cities—with Waymo, Zoox, Tesla, and others scaling operations in San Francisco, Miami, Los Angeles, Phoenix, and beyond—there is no widely adopted, community-facing system to document near-misses, traffic disruptions, unsafe behaviors, or emergency protocol failures in real time.

This PRD defines the mobile application scope for AV Watch's first release (v1.0), grounded in primary qualitative and quantitative user research conducted between February and March 2026. The app will launch as a React Native cross-platform application targeting iOS and Android, with a focus on three core capabilities: structured incident reporting with geolocation and media capture, an interactive incident map for community-level transparency, and aggregated data views that pull from official sources (NHTSA, CA DMV, state databases).

2. Research Insights & Data Analysis

The following insights are synthesized from two primary research sources: a structured Google Forms survey (n=138) distributed to residents of AV-active cities, and qualitative analysis of two high-engagement Reddit threads—a detailed Waymo incident report from r/waymo (519 upvotes, 225+ comments) and a community sentiment thread from r/sanfrancisco (77 comments).

2.1 Survey Demographics & Exposure

The survey captured respondents primarily from the Bay Area (66%), San Francisco (24%), Los Angeles (7%), and Miami (3%), with ages concentrated in the 25–34 range (44%) and 18–24 range (23%). Critically, 73% of respondents encounter AVs at least monthly, and 39% encounter them daily—confirming a population with high AV exposure and direct experiential knowledge.

2.2 Incident Experience Distribution

Incident Type	Experienced 1+	Heard Only	Never	Key Takeaway
AV blocking traffic	36%	36%	28%	Most common issue
Unpredictable driving	23%	45%	32%	High awareness gap
Feeling unsafe	22%	26%	52%	Significant minority
Excessive noise	6%	22%	72%	Low priority
AV-involved injury	1%	48%	51%	Rare but feared

2.3 Reporting Behavior Gap

Perhaps the most striking finding: 83% of respondents have never reported or documented an AV incident, despite 36% having personally experienced at least one. Of those who did report, channels were fragmented—Reddit, Instagram Stories, Twitter/X, direct email to Waymo, and iMessage to friends. Only 7% reported directly to the AV company, and none mentioned any centralized public reporting platform. This gap is the core opportunity AV Watch addresses.

2.4 Qualitative Themes from Reddit Analysis

The Reddit data revealed seven recurring themes that directly inform feature prioritization:

#	Theme	Evidence	Feature Implication
1	**Emergency protocol failures**	Waymo geofence incident: 45-min highway stranding, refusal to release manual control, mismatched emergency response chain	Detailed incident reporting with severity levels, timeline capture, and emergency tagging
2	**Traffic disruption is the #1 lived experience**	Blocking lanes, double parking, freezing at lights, stopping after intersections. Confirmed by survey: 36% experienced it	Quick-report flow for traffic disruptions with drop-pin location and optional photo
3	**Accountability vacuum**	No one knows who to report to. Waymo support is inconsistent. NHTSA form is buried. No feedback loop for reporters	Centralized reporting hub with status tracking and transparent data export
4	**Safety perception varies by road user type**	Pedestrians and cyclists feel overwhelmingly safer. Drivers are frustrated. Passengers are mostly positive but edge cases terrify	Role-based reporting (pedestrian, cyclist, driver, passenger) for nuanced data
5	**Regulatory and jurisdictional confusion**	Debate over who handles stalled AVs: Highway Patrol, Road Rangers, or Waymo? State-level regulatory fragmentation	Policy resource section; incident routing guidance by jurisdiction
6	**Surveillance and data privacy concerns**	Multiple comments about cameras collecting data. Worry about where footage ends up. Children's privacy mentioned	Transparent privacy policy; anonymous reporting option; no account required for basic reports
7	**Economic and transit displacement anxiety**	Jobs lost, public transit underfunded, streets congested by empty repositioning vehicles, no community investment	Data views showing AV fleet density and impact metrics; transit disruption category

2.5 Competitive Landscape

Existing data sources are fragmented and not designed for community participation. The California DMV has received 939 collision reports since 2014, published as individual PDFs with no search, filter, or visualization capability. NHTSA's SGO reporting was recently amended (effective June 2025) to allow 5-day reporting windows, but remains manufacturer-facing only. The Berkeley TIMS AV Safety Dashboard offers collision visualization but excludes near-misses, traffic disruptions, and community reports entirely. Tesla publishes a self-reported Vehicle Safety Report with aggregated

telemetry but no per-incident detail. No existing platform combines official data with community-sourced reports in a mobile-accessible, geolocation-enabled format—this is the whitespace AV Watch occupies.

3. Product Vision & Goals

Vision: AV Watch becomes the default civic tool for autonomous vehicle transparency—a community-powered layer of accountability that gives every road user a voice and every policymaker an evidence base.

3.1 Goals for v1.0 (MVP)

Goal	Description	Success Metric
G1: Report	Enable any road user to submit a structured AV incident report in under 90 seconds from their phone	Avg. report submission time < 90s; 500+ reports in first 3 months
G2: Visualize	Display reported incidents on an interactive, filterable map alongside official data	80% of users who open the map engage with at least one filter
G3: Inform	Provide accessible summaries of AV safety data that are useful to both citizens and policymakers	Monthly data export downloads by at least 5 organizations
G4: Trust	Build a platform that users trust with their data through transparency, anonymity options, and open data principles	< 5% report abandonment rate; > 4.0 app store rating

3.2 Non-Goals for v1.0

- Real-time AV tracking or live vehicle monitoring
- Direct integration with AV company APIs (Waymo, Tesla, etc.)
- Legal claim filing or liability assessment tools
- Social networking features (comments, follows, direct messaging)
- Monetization or advertising of any kind

4. User Personas

4.1 Primary: The Concerned Commuter

Profile: 25–34 year old Bay Area resident who walks, bikes, and uses public transit daily. Encounters AVs multiple times per week. Has experienced at least one traffic disruption but has never formally reported it. Uses Reddit and Instagram.

Pain point: "I’ve seen Waymos block intersections three times this month, but I have no idea who to tell or if it even matters."

Need: A fast, frictionless way to report what they saw, with confidence that the data goes somewhere meaningful.

4.2 Secondary: The Safety-Conscious Rider

Profile: 35–54 year old San Francisco resident who has switched from Uber/Lyft to Waymo. Generally positive about AVs but alarmed by edge-case failures (power outages, geofence errors). Wants AVs to improve.

Pain point: "I support this technology, but when my Waymo froze at a dead traffic light, I had no way to formally document it outside of calling their support line."

Need: A passenger incident report flow that captures in-ride issues with timestamps and severity.

4.3 Tertiary: The Policy Researcher

Profile: Government staffer, academic researcher, or journalist tracking AV deployment impacts. Needs structured, exportable data.

Pain point: "The DMV publishes individual PDF crash reports. NHTSA data is manufacturer-reported. There's no unified view that includes what people actually experience on the street."

Need: Dashboards with filterable data, trend analysis, and CSV/JSON data export.

5. Feature Specifications

5.1 Incident Reporting (P0 – Must Have)

The core of AV Watch. A structured, guided reporting flow optimized for mobile use in real-world conditions (standing on a sidewalk, sitting in an AV, or recalling an incident later).

5.1.1 Report Types & Categories

Category	Sub-types	Data Source
Traffic disruption	Blocking lane, double parking, frozen at intersection, blocking bus/transit, blocking emergency vehicle	Survey (36% experienced), Reddit
Unpredictable driving	Sudden braking, wrong lane, running red/yellow, aggressive maneuver, crossing double lines	Survey (23% experienced), Reddit
Safety concern	Near-miss with pedestrian/cyclist, geofence failure, power outage behavior, construction confusion	Reddit (Miami incident, PG&E outage)
Passenger experience	Unsafe drop-off location, vehicle stuck, support unresponsive, uncomfortable driving style	Survey open-ended, Reddit
Noise/environmental	Excessive noise, idling/repositioning on residential streets, late-night disturbance	Survey (6% experienced)

Collision/injury	Property damage, bodily injury, animal collision	NHTSA/DMV + community reports

5.1.2 Report Flow (4 Steps, < 90 seconds)

- **Step 1 – What happened?** Select incident category from visual cards. One tap. Default to most common (traffic disruption).
- **Step 2 – Where & when?** Auto-fill GPS location and timestamp. User can adjust via map pin drag or manual address entry. Date/time picker for retroactive reports.
- **Step 3 – Details:** Sub-type selection (multi-select chips). AV company selector (Waymo, Zoox, Tesla, Cruise, Other, Unknown). Severity slider (1–5: Minor annoyance → Life-threatening). Your role (pedestrian, cyclist, driver, passenger, bystander). Optional: free-text description (500 char max).
- **Step 4 – Evidence (optional):** Photo/video upload (max 3 files, 30s video). Camera capture directly from app. Submit button with confirmation.

5.1.3 Reporting Principles

- No account required for basic reports (anonymous by default)
- Optional account creation unlocks: report history, edit/update submitted reports, notification on nearby incidents
- Reports are geotagged but exact user location is never stored—only incident location
- All reports receive a unique report ID for reference and follow-up
- Duplicate detection: flag potential duplicates if same location + time + category within 30 minutes

5.2 Interactive Incident Map (P0 – Must Have)

A map-based view that serves as both the primary discovery interface and the public accountability layer.

5.2.1 Map Features

- Heatmap overlay showing incident density by area
- Individual incident pins with color-coded severity (green/yellow/orange/red)
- Cluster view at zoom-out levels; individual pins at street level
- Tap pin to see: category, date, severity, AV company, description snippet, media thumbnail
- Filter panel: by category, AV company, severity, date range, reporter role, data source (community vs. official)
- Toggle layers: community reports only, official data only, or combined

5.2.2 Official Data Integration (v1.0 scope)

- California DMV collision reports (939 reports, scraped/parsed from PDFs into structured database)

- NHTSA Standing General Order incident reports (publicly available)
- Manual data ingestion pipeline with quarterly refresh cadence for v1.0

5.3 Data Dashboard (P1 – Should Have)

Summary views that transform raw incident data into policy-relevant insights.

- Company comparison: incidents per company, by type and severity, over time
- Geographic trends: which neighborhoods/corridors see the most disruption
- Temporal patterns: time-of-day, day-of-week incident distributions
- Community vs. official data comparison: are community reports filling gaps?
- Data export: CSV and JSON download of filtered datasets, with API access planned for v2.0

5.4 Alerts & Notifications (P1 – Should Have)

- Location-based alerts: opt-in notifications when incidents are reported near your frequent locations
- Threshold alerts: notify when a specific area exceeds an incident count in a rolling 7-day window
- Weekly digest: summary email/push with top incidents and trends in your city

5.5 Know Your Rights / Resource Center (P2 – Nice to Have)

Informed by the regulatory confusion surfaced in the Reddit data, this section provides accessible guidance.

- Who to contact if you experience an AV incident (by city/state)
- How to file an official NHTSA complaint (with direct link and step-by-step guide)
- AV operating permits and geofence maps for your city (where available)
- FAQ: Can I take manual control? What happens if an AV hits me? Who is liable?

5.6 AV News & Trends Feed (v2.0 – Post-Launch)

A curated, algorithmically ranked news feed that keeps users informed about AV developments in their city and nationally—without becoming a social media platform. This feature drives return visits and positions AV Watch as a daily-use civic tool, not just a reporting utility.

5.6.1 Content Sources & Ingestion

- RSS/Atom feed aggregation from major outlets covering AV news (e.g., TechCrunch, The Verge, Reuters, local city outlets, NHTSA press releases)
- Official regulatory updates: new NHTSA orders, state DMV permit changes, city council decisions on AV operations
- Platform-native trending incidents: auto-generated cards when report clusters emerge (e.g., “14 traffic disruption reports on Valencia St this week”)

- AV company announcements: expansion into new cities, safety reports, service changes (scraped from company blogs/newsrooms)

5.6.2 Ranking Algorithm

The feed ranking algorithm prioritizes relevance and recency using a weighted scoring model:

Signal	Weight	Description
Geographic relevance	High	Stories about the user's city or state are boosted. Uses device location or user-set home city.
Recency	High	Time-decay function ensures fresh content surfaces first. Breaking regulatory news gets a temporary boost.
Source authority	Medium	Official government sources and established outlets rank above aggregators and blogs.
Topic match	Medium	NLP-based topic extraction matches stories to incident categories the user has reported or viewed.
Engagement signal	Low	"This affects me" reactions from other users provide a lightweight relevance signal without comments.

5.6.3 User Interaction Model

The feed deliberately avoids open-ended comments to maintain data credibility and avoid moderation overhead. Instead, it uses structured engagement:

- **Reaction signals:** "This affects me" and "Important" buttons on news items and trending incident cards. These enrich the ranking algorithm and provide lightweight sentiment data.
- **Save for later:** Bookmark articles to a personal reading list within the app.
- **Share externally:** Native share sheet to push stories to X, Reddit, iMessage, etc.—driving organic growth.
- **Link to report:** Prominent "Report a related incident" CTA on every news item, connecting the news feed back to the core reporting flow.

5.6.4 Technical Approach

- Ingestion pipeline: scheduled workers (cron jobs or cloud functions) scrape RSS feeds, company newsrooms, and regulatory pages every 30–60 minutes
- NLP processing: lightweight topic classification and entity extraction (AV company, city, incident type) using a fine-tuned model or keyword-based rules for MVP
- Deduplication: title similarity matching (cosine similarity on TF-IDF vectors) to prevent the same story appearing from multiple outlets
- Content storage: articles stored as metadata (title, summary, source, URL, published date, extracted entities)—AV Watch links out to original sources rather than hosting full articles, respecting copyright and keeping storage minimal

- Personalization: user's city preference and report history feed into the ranking model; no tracking beyond explicit app interactions

5.6.5 Design Rationale: Why Not Comments?

The decision to exclude open-ended comments is grounded in primary research. Analysis of the Reddit threads used in this study revealed consistent patterns of derailment: ad hominem attacks, accusations of astroturfing, and political debates unrelated to AV safety. A 4-person capstone team cannot build robust content moderation at launch. More importantly, open comments risk undermining AV Watch's positioning as a neutral data platform trusted by policymakers and researchers. Structured reactions preserve engagement value while protecting credibility. Full social features may be reconsidered in a future version if a community governance model is established.

6. Information Architecture

The app uses a bottom tab navigation pattern with four primary screens at launch, expanding to five in v2.0 with the addition of the News tab:

Tab	Screen	Primary Action	Key Components
Map	Incident Map	Explore incidents geographically	Map view, filters, incident pins, heatmap toggle
Report	New Report	Submit an incident	4-step guided flow, camera, GPS, submit
News (v2.0)	AV News & Trends	Stay informed on AV developments	Ranked feed, reaction buttons, save, share, report CTA
Dashboard	Data Insights	Understand trends and patterns	Charts, company cards, date filters, export
Profile	Settings / My Reports	Manage account and report history	Report list, notification prefs, about/privacy

The Report tab is visually prominent (centered, elevated FAB-style button) to encourage reporting as the primary action.

7. Technical Architecture

7.1 Tech Stack

Layer	Technology	Rationale

Mobile client	React Native + Expo	Cross-platform (iOS + Android) from single codebase; rapid iteration; strong community
Maps	Mapbox GL / React Native Maps	Custom styling, heatmap layers, clustering, offline support potential
State management	Zustand or React Context	Lightweight, sufficient for MVP complexity
Backend API	Node.js + Express / FastAPI (Python)	RESTful API; Python option enables direct ML/data pipeline integration
Database	PostgreSQL + PostGIS	Geospatial queries for incident proximity, clustering, and spatial filtering
File storage	AWS S3 / Cloudflare R2	Photo and video upload storage with CDN delivery
Authentication	Firebase Auth (optional accounts)	Social login + email; deferred for anonymous-first flow
Hosting	Vercel (API) + AWS (DB, storage)	Cost-effective for capstone; scalable if adoption grows

7.2 Data Model (Core Entities)

Entity	Key Fields	Type	Constraints	Notes
Incident	id, category, sub_types[], severity, description	Core	category required	Primary report object
Location	latitude, longitude, address, city, state, geohash	Embedded	lat/lng required	PostGIS point geometry
Media	id, incident_id, url, type (photo/video), size	Related	max 3 per incident	S3 presigned upload
Reporter	id, role, anonymous flag, device_id	Optional	device_id for dedup	No PII required
AVCompany	id, name, slug	Reference	Enum set	Waymo, Zoox, Tesla, Cruise, Other, Unknown
OfficialReport	id, source, source_id, date, location, severity, url	Ingested	source required	DMV/NHTSA imports

NewsArticle (v2.0)	id, title, summary, source_url, source_name, published_at, entities[], relevance_score	Ingested	source_url unique	RSS/scrape pipeline; links out to original

8. Design Principles

- **Speed over completeness:** A 3-field report submitted is infinitely more valuable than a 15-field report abandoned. Every optional field must justify its friction.
- **Trust through transparency:** Users can see exactly what data is collected, how it's stored, and who can access it. Privacy policy is written in plain language, not legalese.
- **Neutral platform, not advocacy:** AV Watch documents facts. The app does not editorialize, rank companies, or take positions on whether AVs are good or bad.
- **Accessibility first:** WCAG 2.1 AA compliance. High contrast mode. VoiceOver/TalkBack support. One-handed operation for the full report flow.
- **Data dignity:** Community-submitted data is treated with the same rigor as official sources. Both are labeled clearly. Neither is privileged over the other.

9. Prioritized Roadmap

Phase	Timeline	Deliverables	Success Gate
Phase 1: Core MVP	Mar – Apr 2026	Report flow, incident map, basic filters, anonymous submission, official data import (CA DMV)	50+ test reports submitted; map renders < 2s on 3G
Phase 2: Data Layer	Apr – May 2026	Dashboard views, company comparison, data export (CSV/JSON), NHTSA data integration	Exportable dataset passes validation; 3+ external users download data
Phase 3: Engagement	May 2026 (Capstone)	Push notifications, weekly digest, resource center, onboarding flow, App Store submission	App Store approval; 100+ organic installs in first month
Future: v2.0+	Post-graduation	AV News & Trends feed with algorithmic ranking, public API, ML-based duplicate detection, verified reporter badges, multi-state	News feed DAU/MAU > 40%; community governance model established

		expansion, transit disruption tracking	

10. Risks & Mitigations

Risk	Likelihood	Impact	Mitigation
False/spam reports	High	Erodes data credibility; AV companies dismiss platform	Rate limiting by device ID; community flagging; optional verification for high-severity reports; statistical outlier detection
Low initial adoption	Medium	Insufficient data density for meaningful insights	Launch in SF (highest AV density); partner with cycling/pedestrian advocacy groups; leverage capstone presentation for press
Legal liability	Low	Platform could face claims if reports are inaccurate or defamatory	Clear Terms of Service; reports are user-submitted opinions/observations; no verification claims; consult Berkeley IP counsel
Official data staleness	Medium	DMV/NHTSA data lags by weeks/months	Clearly label data freshness; community data fills real-time gap; automated scraping pipeline for v2.0
Privacy backlash	Medium	Users concerned about being tracked or identified	Anonymous-first design; no account required; GPS used only for incident location, never user tracking; transparent data policy

11. Success Metrics

11.1 North Star Metric

Total verified incident reports submitted per month, segmented by city. This measures both adoption and the platform's core value proposition: giving road users a voice.

11.2 Supporting Metrics

Category	Metric	Target (3 months)
Engagement	Monthly active users (MAU)	1,000+
Engagement	Reports per user per month	> 1.5
Quality	Report completion rate (start to submit)	> 85%
Quality	Median time to complete report	< 90 seconds
Retention	Day 7 retention	> 30%
Retention	Day 30 retention	> 15%
Data	Total reports in database	500+
Data	Geographic coverage (distinct zip codes)	50+
Impact	Data export downloads	10+ unique organizations
Impact	Media mentions / citations	3+

12. Appendix

12.1 Research Data Summary

- Survey instrument: Google Forms, 14 questions, distributed Feb 12–Mar 1, 2026
- Survey sample: n=138, primarily Bay Area / SF / LA / Miami residents
- Reddit source 1: r/waymo – Waymo Miami geofence failure post (519 upvotes, 225+ comments)
- Reddit source 2: r/sanfrancisco – Community sentiment thread (77 comments, 7.4K views)

12.2 Key Regulatory References

- NHTSA Third Amended SGO 2021-01 (effective June 16, 2025): 5-day incident reporting requirement
- California DMV AV collision reports: 939 reports as of Feb 2026
- House Bill H.R. 4376 – AV Safety Data Act (introduced July 15, 2025)
- Berkeley TIMS AV Safety Dashboard: tims.berkeley.edu/tools/avsafety.php

12.3 Open Questions for Team Discussion

- Should v1.0 support web app in addition to mobile, or mobile-only?
- How do we handle incident reports that describe the same event from multiple reporters?
- What is the moderation model for flagged/disputed reports?

- Should we pursue Berkeley IRB approval to use report data in the research paper?
- Partner outreach: SF Bicycle Coalition, Walk SF, SPUR—who leads?